Wide-Scale Sampling with Hard-Argmin Selection and CBF Projection for Safe Quadrotor Navigation in Narrow Passages

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Abstract

Sampling-based NMPC draws candidate controls from a Gaussian centered at a nominal policy, scores them via short rollouts, and selects one for execution. On narrow-passage tasks the conventional pipeline suffers three interacting failure modes: (i) a narrow perturbation scale traps all candidates around a biased nominal; (ii) importance-weighted averaging (MPPI) or iterative refitting (CEM) dilutes the few escape candidates; and (iii) without a safety filter, wide exploration produces collisions. We identify a *three-pillar* coupling that makes test-time scaling effective in non-convex obstacle environments: **(P1)** wide-scale sampling ($\sigma \geq 5\,\mathrm{N}$, per axis), **(P2)** hard-argmin selection to preserve escape quality, and **(P3)** a DT-CCBF safety projection to filter unsafe candidates. P1 and P3 are each individually necessary for success and safety respectively; P2 is necessary for competitive tracking accuracy (soft-weighting degrades TErr by 3–4.4×). The three are jointly sufficient within the narrow-passage quadrotor regime studied here. On a quadrotor narrow-passage benchmark ($N_{\mathrm{MC}} = 80$), TSH-NMPC at $K = 10$ achieves $\geq 93\%$ success versus 63–70% for narrow-scale PD (McNemar $p < 0.001$). Controlled ablations confirm all three pillars: σ-sweep shows success rising from 75% at $\sigma = 1$ to 100% at $\sigma \geq 5$ (P1); MPPI/CEM at matched $\sigma = 5$ reach comparable success but 3–4.4× worse TErr than hard-argmin (P2); disabling CBF yields 40/40 collisions (P3). Under +50% mass mismatch, PD collapses to 0% while wide-scale variants retain 78–83% success. A diversity-over-accuracy ablation replacing the random source with a learned predictor finds the random source superior ($N_{\mathrm{MC}} = 300$, $p = 7.4 \times 10^{-6}$), identifying diversity as the driving mechanism of P1. Against CasADi+IPOPT ($H = 20$), the controller achieves success-rate parity (McNemar $p = 1.0$) with approximately deterministic compute budget (no NLP-solver dependency), although CasADi retains a 5–11× advantage in terminal error (0.003 vs. 0.015–0.032 m at $K \geq 20$) and lower mean latency.

Keywords: Nonlinear model predictive control; sampling-based MPC; control barrier functions; quadrotor obstacle avoidance; ablation study.

1. Introduction

Sampling-based nonlinear model predictive control (NMPC) — MPPI [13], the Cross-Entropy Method (CEM) [2], and their many variants — has become a standard tool for online obstacle avoidance on quadrotors and ground vehicles. Its appeal is twofold. First, the rollout-and-score evaluation imposes no convexity or differentiability requirement on the stage cost: any executable obstacle model is acceptable. Second, the per-step compute is bounded — K short rollouts, each of horizon S — and parallelizes trivially on commodity hardware.

The standard formulation, however, is *single-source*: at each control instant, K candidate first-step controls are drawn from one Gaussian distribution centered at a nominal control (the previous optimal, a warm-start PD policy, or the zero command). On many navigation tasks this single source is sufficient. But on tight passages — where the mission requires threading a gap between obstacles, rather than detouring around — the nominal control often points *into* an obstacle and every candidate inherits the bias. Increasing K produces diminishing returns: more samples around a bad nominal do not escape it. Our experiments confirm this brittleness — single-source PD-Gaussian rollout NMPC achieves only 73.8% success at $K = 20$ on a narrow-passage quadrotor benchmark (Section 5.2).

This paper identifies a *three-pillar* mechanism that makes test-time scaling effective in non-convex obstacle environments: **(P1)** wide-scale perturbation ($\sigma \geq 5\,\mathrm{N}$) to escape the biased nominal; **(P2)** hard-argmin selection (not importance-weighted averaging) to preserve escape-candidate quality; and **(P3)** DT-CCBF safety projection to filter unsafe wide-scale candidates. P1 and P3 are individually necessary for success and safety respectively; P2 is necessary for competitive tracking accuracy. The three are jointly sufficient within the narrow-passage regime (Section 5.5.3). The proposed controller instantiates all three: K candidates drawn from a PD-Gaussian at $\sigma = 5$, scored by short rollout, hard-argmin selected, and CBF-projected. We present the wide-scale sampler in two equivalent forms: $R0\_pd\_s5$ (single-source-wide at $\sigma = 5$) and $R1\_s5$ (two-source 50/50 mixture of $\sigma_{\mathrm{pd}} = 2$ and $\sigma_r = 5$). Paired McNemar finds them equivalent ($p = 0.51$, Section 5.5.2), confirming the lever is the noise scale, not the source partition. The resulting controller is called TSH-NMPC — Test-time Scaling with Hard-argmin.

1.1 Contributions

1. **A three-pillar mechanism for effective test-time scaling in non-convex obstacle environments.** Wide-scale sampling ($\sigma \geq 5$), hard-argmin selection, and DT-CCBF safety projection are jointly sufficient (Section 5.5.3): removing P1 collapses success to $\leq 75\%$; replacing P2 with soft-weighting (MPPI) degrades TErr by $4.4\times$; removing P3 yields 40/40 collisions. Under +50% mass the three-pillar controller retains $\geq 77.5\%$ success where PD collapses to 0/40 (Section 5.5).

2. **A rollout-based selector with monotone K-improvement** (Proposition 3) at mean 6.2 ms per step at $K = 10$ on commodity CPU (Section 5.10).

3. **A diversity-over-accuracy ablation.** Replacing the random wide source by a 27,935-parameter learned predictor is significantly *outperformed* by the random source at $N_{\mathrm{MC}} = 300$ (McNemar $p = 7.4 \times 10^{-6}$, Section 5.6): candidate *diversity* beats candidate *accuracy* when escape geometry is approximately isotropic.

4. **Success-rate parity with CasADi+IPOPT at approximately deterministic compute budget and no NLP dependency.** Paired McNemar $p = 1.0$ on `narrow` $N_{\mathrm{MC}} = 80$ at $K \geq 20$; CasADi retains a 5–11$\times$ TErr advantage and lower mean latency, but its per-step latency is data-dependent (max = 35 ms) whereas TSH-NMPC's compute is $\mathcal{O}(K \cdot S)$ and algorithmically data-independent (CV < 20%) (Section 5.7).

1.2 Paper Organization

Section 2: dynamics and background. Section 3: TSH-NMPC controller and rollout selector. Section 4: CBF forward-invariance and monotone min-cost. Section 5: experiments across three task families. Section 6: mechanism, scope, limitations. Section 7: conclusion.

1.3 Related Work

Sampling NMPI/CEM. MPPI [13, 14] computes importance-weighted averaging of K Gaussian-perturbed trajectories; this is a strength on race-track tasks but a liability in tight passages [4, 9]. CEM [2, 11] iteratively refits to the top-β quantile. TSH-NMPC is a one-shot wide-scale scheme: diversity is paid once in a fixed-budget pool rather than per-iteration (equivalently, zero-iteration CEM with fixed $\sigma_r^2 I$).

Control barrier functions. Continuous-time CBFs [1] enforce safe-set forward-invariance; discrete-time variants [12] extend the guarantee under sampled-data execution. We use the off-the-shelf DT-CCBF as the safety filter.

Learned warm-starts. Several works replace the Gaussian by a learned proposal [7, 8, 10, 21], but rarely compare against an isotropic-random source at matched budget. Section 5.6 contributes such a comparison.

Narrow-passage planning & diversity. The narrow-passage difficulty [3, 6] is typically attacked at the planner level; our controller-level analogue shifts diversity to the inner control loop. The exploitation–exploration tension has analogues in MCTS [16], progressive widening [17], evolution strategies [18, 19], and best-arm identification [20].

Scenario optimization. Randomized optimization [24, 25] studies the feasibility and cost of solutions drawn from finitely many random scenarios: as the sample count K grows, the violation probability decays at a known rate. TSH-NMPC's hard-argmin over K rollout-evaluated candidates is an instance of this paradigm—*without* the convexity assumption—and Proposition 3 provides the corresponding monotonicity guarantee (Section 4).

2. Problem Formulation and Background

2.1 Quadrotor Abstract Translational Dynamics

Following standard quadrotor cascade control, the outer-loop translational guidance operates on a 6D state $x_t = [p_t^\top, v_t^\top]^\top \in \mathbb{R}^6$, where $p_t \in \mathbb{R}^3$ is position and $v_t \in \mathbb{R}^3$ is velocity. The virtual control $u_t \in \mathbb{R}^3$ is the gravity-precompensated net force command; the inner-loop attitude controller is assumed fast enough ($T_{\text{att}} \ll \Delta t$) to track the flatness-mapped thrust and attitude without affecting the translational closed-loop dynamics. The abstract discrete-time dynamics with time step Δt are

$$x_{t+1} = A_d x_t + B_d u_t, \quad A_d = \begin{pmatrix} I_3 & \Delta t\, I_3 \\ 0 & I_3 \end{pmatrix},\; B_d = \tfrac{\Delta t}{m} \begin{pmatrix} 0 \\ I_3 \end{pmatrix}, \tag{1}$$

where m is the quadrotor mass and body drag is omitted in the nominal model for brevity (drag appears in the simulation environment of Section 5.1). Input saturation is $u_t \in [u_{\min}, u_{\max}]^3$ with $u_{\max} = 15\,\text{N}$.

2.2 Discrete-Time Composite CBF (DT-CCBF)

Let $\mathcal{C}_i = \{x \in \mathbb{R}^6 : h_i(x) \geq 0\}$ be the obstacle-free set for obstacle i, with smooth barrier function $h_i(x) = \|p - p_i\|^2 - r_i^2$. The unsafe set $\cup_i \overline{\mathcal{C}}_i$ is non-convex. We replace the minimum of per-obstacle barriers by the smooth log-sum-exponential composite

$$h(x) = -\tfrac{1}{\kappa} \log \sum_{i=1}^{M} \exp\big(-\kappa\, h_i(x)\big), \quad \kappa = 5, \tag{2}$$

with $h(x) \leq \min_i h_i(x)$ and $h(x) \to \min_i h_i(x)$ as $\kappa \to \infty$.

Remark (LSE conservatism and narrow passages). For $M = 5$ obstacles and $\kappa = 5$, the LSE gap $h(x) \le \min_i h_i(x)$ satisfies $\min_i h_i(x) - h(x) \le \kappa^{-1} \log M \approx 0.32\,\mathrm{m}$. This gap is a *uniform* lower bound: the composite barrier perceives obstacles as $\approx$0.32 m larger in radius than they physically are. On the `narrow` benchmark, where the gap between obstacles is $\approx$0.6 m (a 4:1 ratio to the 0.15 m safety radius), the LSE barrier sees a channel width of $\approx$0.28 m—less than half the geometric width. This conservatism is one factor contributing to single-source PD's poor narrow-passage success (66% at $K = 10$): the PD nominal u_{pd} points toward a feasible geometric path that the *conservative composite barrier* treats as unsafe, creating a structural bias that narrow $\sigma = 2$ cannot escape. Wide-scale sampling ($\sigma = 5$) partially overcomes this by generating candidates aggressive enough to thread the *geometric* gap while the CBF projects them onto the (conservative) safe set. The discrete-time CBF condition [1, 12]

$$h(x_{t+1}) \ge (1 - \alpha_d)\, h(x_t) - \gamma_d \tag{3}$$

is enforced with $(\alpha_d, \gamma_d) = (0.8, 0.2)$. Linearizing (3) about the nominal dynamics (1) yields the relative-degree-2 CBF Jacobian and the QP constraint $A(x_t)^\top u \le b(x_t)$. The safety filter solves

$$u_{\mathrm{safe}} = \operatorname*{arg\,min}_{u \in [u_{\min}, u_{\max}]^3} \tfrac{1}{2}\|u - u_{\mathrm{nom}}\|^2 \quad \text{s.t. } A(x_t)^\top u \le b(x_t). \tag{4}$$

When (4) is infeasible, a box-constrained emergency fallback with hard clearance $\delta_{\mathrm{buffer}} = 0.15\,\mathrm{m}$ is used.

2.3 Sampling NMPC

Given a state x_t, a setpoint x_{sp}, and a candidate budget K, a *sampling NMPC controller* builds K first-step control candidates $\{u^{(k)}\}_{k=1}^{K}$, evaluates each by a simulation-based cost $J^{(k)}$, and executes the first-step control of the lowest-cost candidate after projecting it onto the CBF safety set. The controller is called *single-source* when all K candidates are drawn from a single Gaussian distribution

$$u^{(k)} \sim \mathcal{N}(\mu_t, \sigma^2 I_3), \quad k = 1, \ldots, K, \tag{5}$$

centered at some nominal μ_t (the previous optimal, a warm-start policy, or zero). MPPI [13] replaces the hard argmin by an importance-weighted average and inherits the single-source structure; CEM [2] iteratively refits the parameters of a single-source Gaussian to the top-β quantile across several iterations. The controller is called *two-source* in Section 3 when the pool comprises two Gaussians with different covariances at the same nominal.

3. Method: Wide-Scale Sampling with Hard-Argmin Selection and CBF Projection

This section presents TSH-NMPC (Test-time Scaling with Hard-argmin): K candidates from a wide-scale PD-centered Gaussian, evaluated by short rollout, hard-argmin selected, and CBF-projected. No learned predictor, importance weighting, or iterative refit is used (a learned-prior ablation appears in Section 5.6).

3.1 Wide-Scale PD-Centered Candidate Generation

At each closed-loop step t, given the current state $x_t \in \mathbb{R}^6$, the setpoint $x_{\mathrm{sp}} \in \mathbb{R}^6$, and the candidate budget $K \in \mathbb{N}_+$, the controller forms a pool of K first-step controls $\{u^{(k)}\}_{k=1}^{K} \subset \mathbb{R}^3$ from a wide-scale PD-centered Gaussian.

Default: single-source-wide (*R0_pd_s5*). Let $u_{\mathrm{pd}}(x_t, x_{\mathrm{sp}}) = m(K_p(p_{\mathrm{sp}} - p_t) + K_d(v_{\mathrm{sp}} - v_t))$ denote the model-aware PD first-step control with gains $(K_p, K_d) = (4, 3)$, where m is the (nominal) quadrotor mass. The default pool draws all K candidates from a single wide Gaussian:

$$u^{(k)} \sim \mathcal{N}(u_{\mathrm{pd}}, \sigma_r^2 I_3), \quad k = 1, \ldots, K, \tag{6}$$

with $\sigma_r = 5.0\,\mathrm{N}$ (per axis; range $[5, 8]$ validated in Section 5.5.3). At $\sigma_r \approx u_{\max}/3$, this distribution is wide enough to escape PD basins that point into obstacles while remaining narrow enough for the rollout selector to identify high-quality candidates.

Two-source variant (R1_s5). An alternative splits the budget into $K_A = \lfloor K/2 \rfloor$ tight samples ($\sigma_{\mathrm{pd}} = 2.0$) and $K_B = K - K_A$ wide samples ($\sigma_r = 5.0$):

$$u^{(k)} \sim \mathcal{N}(u_{\mathrm{pd}}, \sigma_{\mathrm{pd}}^2 I_3),\ k = 1, \ldots, K_A; \quad u^{(k)} \sim \mathcal{N}(u_{\mathrm{pd}}, \sigma_r^2 I_3),\ k = K_A{+}1, \ldots, K. \tag{7}$$

Paired McNemar shows this is statistically indistinguishable from the single-source-wide default ($p = 0.51$, Section 5.5.2); the lever is σ_r, not the source partition. Retained for backwards compatibility only.

All candidates are clamped element-wise to the actuator bounds:

$$u^{(k)} \leftarrow \mathrm{clip}(u^{(k)}, u_{\min}, u_{\max}), \quad k = 1, \ldots, K. \tag{8}$$

3.2 Rollout-Based First-Step Cost Evaluation

The K candidates are scored by a short closed-loop simulation under the nominal model. For each $u^{(k)}$, the system is initialized at x_t, the first step applies $u^{(k)}$, and the remaining $S - 1$ steps apply the PD baseline:

$$\tilde{u}_s^{(k)} = \begin{cases} u^{(k)}, & s = 0, \\ u_{\mathrm{pd}}(\tilde{x}_s^{(k)}, x_{\mathrm{sp}}), & 1 \le s \le S-1, \end{cases} \tag{9}$$

with $\tilde{x}_{s+1}^{(k)} = f(\tilde{x}_s^{(k)}, \tilde{u}_s^{(k)})$ for $s = 0, \ldots, S-1$, where f is the discrete-time abstract translational dynamics from Section 2. We use $S = 20$ (a $0.4\,\mathrm{s}$ horizon).

The candidate cost is the standard NMPC quadratic with an obstacle penalty:

$$J^{(k)} = \sum_{s=1}^{S} \left(\|\tilde{x}_s^{(k)} - x_{\mathrm{sp}}\|_Q^2 + R\,\|\tilde{u}_s^{(k)}\|^2 + \rho \sum_i [\delta - d_i(\tilde{x}_s^{(k)})]_+^2 \right), \tag{10}$$

with $Q = \mathrm{diag}(15, 15, 15, 1, 1, 1)$, $R = 0.02$, $\rho = 2000$, $\delta = 0.3\,\mathrm{m}$, and $d_i(x) = \|p - p_i\| - r_i$. The nominal first-step control is selected by argmin:

$$k^\star = \operatorname*{arg\,min}_{1 \le k \le K} J^{(k)}, \quad u_{\mathrm{nom}} = u^{(k^\star)}. \tag{11}$$

The rollout evaluator (9)–(10) replaces both Q-head pre-screening and MPPI importance-weighted averaging. It estimates the PD-warm-started cost-to-go by averaging over K candidates and an S-step rollout; the monotone expected-cost property (Proposition 3) provides the formal relationship between budget K and selected cost.

Remark (PD warm-start bias). Under +50% mass the PD command is undersized, but this bias is *shared* across all K candidates; hard-argmin operates on the ordinal ranking, not absolute cost, and is therefore robust to a shared additive bias in the rollout evaluator.

3.3 DT-CCBF Safety Projection

The nominal control u_{nom} from (11) is projected onto the safe set via the QP (4), solved in closed form via active-set on $\mathbb{R}^3$. The resulting u_{safe} is executed.

Algorithm 1 TSH-NMPC: one closed-loop step at state x_t

Require: state x_t, setpoint x_{sp}, budget K, scale σ_r, horizon S

1: $u_{\text{pd}} \leftarrow m(K_p(p_{\text{sp}} - p_t) + K_d(v_{\text{sp}} - v_t))$ ▷ PD nominal
2: sample $u^{(1:K)} \sim \mathcal{N}(u_{\text{pd}}, \sigma_r^2 I)$ ▷ wide-scale PD-centered
3: clip $u^{(1:K)}$ to $[u_{\min}, u_{\max}]$
4: **for** $k = 1, \ldots, K$ **(vectorized) do**
5: simulate S-step rollout (9); compute $J^{(k)}$ via (10)
6: $u_{\text{nom}} \leftarrow u^{(\arg\min_k J^{(k)})}$ ▷ hard-argmin
7: build $(A(x_t), b(x_t))$ from DT-CCBF (3)
8: $u_{\text{safe}} \leftarrow$ solve QP (4); fallback if infeasible
9: **return** u_{safe}

Table 1. TSH-NMPC hyperparameters (defaults used throughout Section 5).

Symbol	Default	Role
K	10	total candidate budget
σ_r	5.0 N	wide-scale noise (per axis; range $[5, 8]$)
σ_{pd}	2.0 N	tight-scale noise (legacy; two-source variant only, equivalent to single-source-wide per Sec. 5.5.2)
S	20 (= 0.4 s)	rollout horizon
κ	5	DT-CCBF LSE smoothness

3.4 Algorithm

Algorithm 1 summarizes a single closed-loop step.
Two-source variant: replace Line 3 with $K_A \leftarrow \lfloor K/2 \rfloor$ tight samples at $\sigma_{\text{pd}} = 2$ plus $K_B = K - K_A$ wide samples at σ_r; see (7). Empirically equivalent to the default single-source-wide form (Section 5.5.2).

3.5 Hyperparameter Summary

The proposed controller has five hyperparameters beyond the dynamics (Table 1). All experiments use these defaults unless explicitly swept.

4. Theoretical Properties

This section records two theoretical properties of TSH-NMPC. We intentionally keep the theory minimal: the proposed controller is a sampling scheme on top of an off-the-shelf DT-CCBF, and we make no claim of stability or convergence beyond what either component already provides.

4.1 Forward-Invariance of the Safe Set

Relative-degree-2 derivation. Because the per-obstacle barrier $h_i(x) = \|p - p_i\|^2 - r_i^2$ depends only on position p and the input u enters acceleration through (1) ($\dot{v} = u/m$), the input has relative degree 2 with respect to h_i. Expanding $h(x_{t+1})$ along the Euler-approximated update gives

$$h(x_{t+1}) \;\approx\; h(x_t) + \nabla_p h^\top \big(\Delta t\, v_t + \tfrac{1}{2}(\Delta t)^2 u/m\big) + \tfrac{1}{2} v_t^\top \nabla_p^2 h\, v_t\, (\Delta t)^2, \tag{12}$$

which is affine in u. Substituting into (3) yields $A(x_t)^\top u \le b(x_t)$ with $A(x_t) = -\frac{1}{2}(\Delta t)^2 m^{-1} \nabla_p h(x_t)$ and $b(x_t) = \alpha_d\, h(x_t) + \gamma_d + \nabla_p h^\top (\Delta t\, v_t) + \frac{1}{2}(\Delta t)^2 v_t^\top \nabla_p^2 h\, v_t$. The gradient $\nabla_p h$ is the softmin-weighted $\nabla_p h(x) = \sum_i w_i(x)\, \nabla_p h_i(x)$ with $w_i(x) = \exp(-\kappa h_i)/\sum_j \exp(-\kappa h_j)$.

LSE smoothing margin. The LSE composite is a uniform lower bound: $h(x) \leq \min_i h_i(x)$ with gap $\leq \kappa^{-1} \log M \approx 0.32\,\mathrm{m}$ (M=5, κ=5). Since the gap is conservative, $h(x) \geq 0$ implies $\min_i h_i(x) \geq 0$.

Proposition 1 (Practical safety, conditional on QP feasibility)**.** *Let* $\tilde{h}(x) := h(x) + \gamma_d/\alpha_d$ *and* $\tilde{\mathcal{C}} := \{x : \tilde{h}(x) \geq 0\}$*. Conditional on QP* (4) *feasibility at every step,* $\tilde{\mathcal{C}}$ *is forward invariant under the nominal dynamics:* $x_t \in \tilde{\mathcal{C}} \Rightarrow x_{t+1} \in \tilde{\mathcal{C}}$*. The executed trajectory remains in* $\{x : h(x) \geq -\gamma_d/\alpha_d\}$ *(margin = 0.25 m under our parameters). The proposition does* not *apply when the QP is infeasible (Section 5.8).*

Proof sketch. QP feasibility gives $h(x_{t+1}) \geq (1-\alpha_d)h(x_t) - \gamma_d$. Adding γ_d/α_d yields $\tilde{h}(x_{t+1}) \geq (1-\alpha_d)\tilde{h}(x_t)$, so $\tilde{h} \geq 0$ is preserved. The LSE margin ($\leq 0.32\,\mathrm{m}$) plus the dynamic margin (0.25 m) give a worst-case true-barrier margin of $\approx$0.57 m. See [12, Thm. 1], [1, §IV]. □

Remark (Fallback and feasibility). When the QP is infeasible due to actuator saturation, the box-constrained emergency fallback $u_{\text{safe}} = \operatorname{sgn}(-A(x_t)) \odot u_{\max}$ (element-wise sign, applying full actuator effort in the barrier-gradient direction) minimizes the one-step constraint violation but does *not* preserve $\tilde{\mathcal{C}}$. Section 5.8 reports the empirical fallback rate; $\delta_{\text{buffer}} = 0.15\,\mathrm{m}$ provides headroom for the one-step excursions observed in practice.

Proposition 2 (One-step fallback excursion bound)**.** *Suppose the QP* (4) *is infeasible at step* t *and the box-constrained fallback* $u_{\text{fb}} = \operatorname{sgn}(-A(x_t)) \odot u_{\max}$ *(element-wise sign) is executed. Then the one-step decrease of the smooth composite barrier satisfies*

$$h(x_t) - h(x_{t+1}) \;\leq\; \Delta t\, \|\nabla_p h(x_t)\|\, \|v_t\| \;+\; \tfrac{1}{2}(\Delta t)^2 m^{-1} u_{\max}\, \|\nabla_p h(x_t)\|_1 \;+\; O((\Delta t)^3), \tag{13}$$

where $\|\nabla_p h\|$ *is the softmin-weighted gradient norm and* $\|\nabla_p h\|_1$ *is its* ℓ_1 *norm.*

Proof sketch. Apply (12): the drift contributes at most $\Delta t\, \|\nabla_p h\|\, \|v_t\|$ (Cauchy–Schwarz), and the input term at most $\frac{1}{2}(\Delta t)^2 m^{-1} u_{\max}\, \|\nabla_p h\|_1$. The realized excursion is dominated by the velocity drift ($\approx$0.06 m per step), which $\delta_{\text{buffer}} = 0.15\,\mathrm{m}$ absorbs under $v_{\max} \approx 3\,\mathrm{m/s}$. □

Three-layer safety architecture. Proposition 1 provides forward-invariance *conditional on QP feasibility*—the ideal case. Proposition 2 bounds the one-step excursion when feasibility fails—the worst case. The geometric buffer $\delta_{\text{buffer}} = 0.15\,\mathrm{m}$ absorbs the gap between them: under typical velocities ($v \leq 3\,\mathrm{m/s}$), the drift-dominated excursion ($\approx$0.06 m per infeasible step) is well within the buffer, so isolated infeasibilities are tolerated without collision. The three layers (formal invariance, bounded excursion, geometric buffer) together provide the practical safety observed empirically: zero collisions across all trials with CBF enabled (Section 5.8), even at 5–17% fallback rates (Section 5.8).

Consecutive-streak scope. Proposition 2 bounds a *single* infeasible step; streaks compound the drift. Empirically, fallback rates are 5–17% with zero collisions on successful trials (Section 5.8).

4.2 Monotone Expected Min-Cost in K

Proposition 3 (Monotone expected cost)**.** *Let* $K_A = \lfloor K/2 \rfloor$*,* $K_B = K - K_A$*, and let* $u^{(1)}, \ldots, u^{(K_A)} \sim \mathcal{N}(\mu, \sigma_{\text{pd}}^2 I)$ *and* $u^{(K_A+1)}, \ldots, u^{(K)} \sim \mathcal{N}(\mu, \sigma_r^2 I)$ *be drawn independently from each source, then clamped to* $[u_{\min}, u_{\max}]^3$*. Let* $J(\cdot)$ *be the rollout cost* (10) *and define* $J_K^\star := \min_{1 \leq k \leq K} J(u^{(k)})$*. Then* $\mathbb{E}[J_{K+1}^\star] \leq \mathbb{E}[J_K^\star]$ *for every* $K \geq 1$*.*

Proof. Let F_A and F_B be the CDFs of $J(u^{(k)})$ under Source A and Source B respectively. For stratified allocation, the CDF of $J_K^\star$ is $1 - (1-F_A)^{K_A}(1-F_B)^{K_B}$, which is pointwise non-decreasing in both K_A and K_B. The i.i.d. argument of Proposition 2 in the earlier draft is a special case; the same stochastic ordering holds for the stratified pool. □

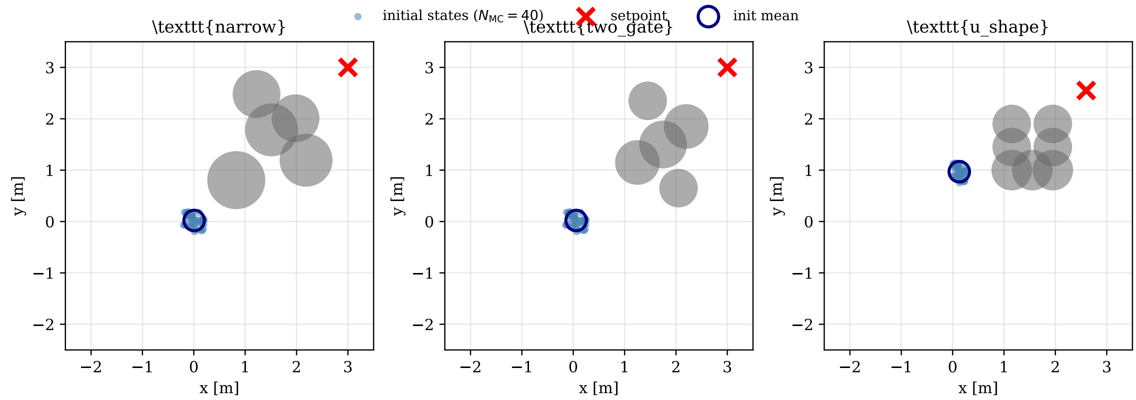

Figure 1. Top-down (xy) projection of the three benchmark tasks. Gray disks: obstacles; red $\times$: setpoint; blue dots: initial positions. `narrow`: five-obstacle gap; `two_gate`: homotopy choice; `u_shape`: concave trap.

Remark. Proposition 3 is the non-convex analogue of the monotone-cost property in scenario optimization [24, 25]: drawing more scenarios cannot worsen the expected selected cost. It establishes non-pathology only; it does *not* claim a rate, nor does it claim convergence to the true optimum of the continuous control space.

5. Experiments

We evaluate TSH-NMPC against a single-source PD-rollout baseline on three quadrotor obstacle-avoidance task families, sweep the Source-B scale σ_r, test under perturbed quadrotor parameters, run a a diversity-over-accuracy ablation that replaces the random second source with a learned-prior second source, and report wall-clock latency. All numerical results below come from the raw JSON files in `experiments/results_v6/`; no figure or table is hand-edited.

5.1 Setup

Tasks. Three task families (Fig. 1). *Narrow* places five spherical obstacles forming tight passages; the setpoint requires threading the gap. *Two-gate* presents a homotopy choice between two equally short corridors. *U-shape* uses a concave layout that traps local-descent controllers. All use the dynamics from Section 2 with $m = 1.5\,\mathrm{kg}$, $b = 0.1$, $\Delta t = 0.02\,\mathrm{s}$, $u_{\max} = 15\,\mathrm{N}$, $N_{\text{steps}} = 150$ (3 s), and CBF parameters $(\alpha_d, \gamma_d, \kappa) = (0.8, 0.2, 5)$. Obstacle positions receive a small random perturbation $\delta_p \sim \mathcal{U}(-0.03, +0.03)^3\,\mathrm{m}$ per trial to prevent overfitting to a fixed geometry.

Quadrotor configurations. Three nominal-mass/drag settings: *nominal* $(m, b) = (1.5, 0.1)$, +50% *mass* $(2.25, 0.1)$, and +50% *drag* $(1.5, 0.15)$. Controllers are *not* re-tuned across configurations.

Methods. `R1_s`σ_r: TSH-NMPC with Source B = random-around-PD at scale σ_r. `R2_s`σ_r: Source B = random-around-zero (worst-case control, Section 5.6 only). `PD`: single-source $\mathcal{N}(u_{\text{pd}}, \sigma_{\text{pd}}^2 I)$ of budget K. `A3`: learned-prior Source B (Section 5.6). Main result sweeps $K \in \{5, 10, 20\}$; elsewhere $K = 10$.

Metrics. *Success*: TErr $\leq 0.3\,\mathrm{m}$ *and* collision-free. *TErr*: $\|p_T - p_{\text{sp}}\|$ on successful trials. *IAE*: $\sum_t \|p_t - p_{\text{sp}}\|\,\Delta t$ on successful trials. *Latency*: per-step wall-clock, single-threaded x86.

Statistical procedure. Each (task, method, K) cell is run for N_{MC} paired Monte Carlo trials. Paired comparisons report (i) McNemar exact two-sided p-value on discordant counts (b, c), and (ii) bootstrap 95% CIs on ΔTErr and ΔIAE over *all* N_{MC} trials; failed trials receive a large penalty (TErr $= 10\,\mathrm{m}$, IAE $= 20$), so the estimator marginalizes over both tracking

Table 2. Main result on `narrow` ($N_{\mathrm{MC}} = 80$ paired trials). Bold = significant ($p < 0.05$, CI excludes zero). $\Delta = \mathrm{R1\_s5} - \mathrm{PD}$ is the marginal estimator (failures imputed TErr = 10 m, IAE = 20).

K	Method	succ. (Wilson 95% CI)	TErr [m] (mean, terminal)	IAE (mean, terminal)	McNemar (b, c)	ΔTErr [95% CI]	ΔIAE [95% CI]
5	PD	46/80 = 57.5% [46.4, 67.9]	0.644	9.31	—	—	—
5	R1_s5	70/80 = 87.5% [78.5, 93.0]	0.171	6.94	$\mathbf{(3, 27)}$, $\mathbf{p < 0.001}$	$\mathbf{-3.04\,[-4.26, -1.92]}$	$\mathbf{-5.11\,[-6.51, -3.72]}$
10	PD	53/80 = 66.3% [55.4, 75.6]	0.380	8.53	—	—	—
10	R1_s5	79/80 = 98.8% [93.2, 99.8]	0.039	6.22	$\mathbf{(1, 27)}$, $\mathbf{p < 0.001}$	$\mathbf{-3.29\,[-4.40, -2.29]}$	$\mathbf{-5.64\,[-6.95, -4.37]}$
20	PD	59/80 = 73.8% [63.1, 82.2]	0.273	8.03	—	—	—
20	R1_s5	77/80 = 96.3% [89.6, 98.7]	0.040	5.82	$\mathbf{(1, 19)}$, $\mathbf{p < 0.001}$	$\mathbf{-2.29\,[-3.28, -1.30]}$	$\mathbf{-4.54\,[-5.75, -3.37]}$

Table 3. Cross-task no-regression ($K = 10$, $N_{\mathrm{MC}} = 40$). $\Delta = \mathrm{R1\_s5} - \mathrm{PD}$; all four CIs exclude zero (R1_s5 lower on both metrics, both tasks).

Task	Method	succ.	TErr [m]	IAE	Δ (R1_s5−PD), 95% CI
two_gate	PD	40/40	0.031	4.31	—
two_gate	R1_s5	40/40	**0.013**	**4.03**	ΔTErr = −0.019 [−0.022, −0.015]; ΔIAE = −0.277 [−0.329, −0.224]
u_shape	PD	40/40	0.027	3.39	—
u_shape	R1_s5	40/40	**0.014**	**3.09**	ΔTErr = −0.013 [−0.016, −0.010]; ΔIAE = −0.303 [−0.351, −0.251]

quality and success/failure. The sign of every Δ is invariant across a 10× sweep of the penalty constants. Wilson 95% CIs are reported for individual success rates. Significance is claimed when $p < 0.05$ *and* the CI excludes zero.

Multiple-comparison correction. Reported p-values are raw paired McNemar exact tests. Holm–Bonferroni [5] at $m = 13$ confirms 7/13 tests at family-corrected level; sensitivity at $m = 28$ yields the same qualitative conclusions. Details in Appendix G.

5.2 Main Result: Narrow Benchmark, $N_{\mathrm{MC}} = 80$

Table 2 reports the main comparison on `narrow` with $N_{\mathrm{MC}} = 80$ paired trials and $K \in \{5, 10, 20\}$. TSH-NMPC (`R1_s5`) achieves a measurable advantage over single-source PD on every K tested: higher success rate, lower TErr, and lower IAE, all with McNemar $p < 0.001$ and bootstrap CIs excluding zero. At $K = 10$, success jumps from 66.3% (PD) to 98.8% (`R1_s5`), and TErr drops by 9.8×.

Two observations: (i) increasing K in single-source PD raises success from 57.5% to only 73.8% — a 16-point gap to TSH-NMPC at $K = 10$; (ii) `R1_s5` at $K = 5$ already beats PD at $K = 20$ (87.5% vs 73.8%). The advantage is *qualitative*, not a sample-efficiency improvement.

Remark 1 (Seed-attribution disclosure)**.** *Main results use seed* 2026*; a seed-*7777 *replication gives PD K*= 10 *at* 56–51/80 *depending on controller-list ordering. Initial-condition pairing is preserved within each file.*

5.3 Cross-Task Generalization

On `two_gate` and `u_shape` ($K = 10$, $N_{\mathrm{MC}} = 40$), PD achieves 100% success; the question is *no regression.* Table 3 confirms R1_s5 matches PD on success and strictly improves TErr and IAE.

5.4 σ_r Sweep: Source-B Operating Range

We sweep $\sigma_r \in \{1, 2, 3, 5, 8\}$ at $K = 10$, $N_{\mathrm{MC}} = 40$ on `narrow` to identify the Source-B scale where the two-source advantage is realized. Table 4 and Fig. 3 show three regimes:

(i) $\sigma_r \le 1\,\mathrm{N}$: Source B too tight, no advantage. (ii) $\sigma_r \in [2, 3]\,\mathrm{N}$: monotone improvement on continuous metrics but success rate below saturation. (iii) $\sigma_r \ge 5\,\mathrm{N}$: success saturates at 100%, TErr and IAE continue to improve monotonically. We label $\sigma_r \in [5, 8]$ the *operating range.*

Table 4. σ_r sweep on `narrow` ($K = 10$, $N_{\mathrm{MC}} = 40$). PD baseline: 27/40, TErr = 0.294, IAE = 8.52.

σ_r [N]	succ.	TErr [m] (succ.)	IAE (succ.)	McNemar (b, c), p vs PD	ΔTErr, ΔIAE vs PD, 95% CI
1	$30/40 = 75.0\%$	0.261	8.00	$(5, 2)$, 0.453 (n.s.)	-0.043 $[-0.070, -0.014]$; -0.430 $[-0.649, -0.227]$
2	$36/40 = 90.0\%$	0.236	7.54	$(9, 0)$, **0.004**	-0.065 $[-0.092, -0.040]$; -0.874 $[-1.071, -0.672]$
3	$34/40 = 85.0\%$	0.195	7.28	$(9, 2)$, 0.065 (marg.)	-0.065 $[-0.094, -0.039]$; -1.169 $[-1.399, -0.947]$
5	$\mathbf{40/40 = 100\%}$	**0.036**	**6.23**	$(13, 0)$, $<$ **0.001**	$\mathbf{-0.094}$ $[\mathbf{-0.125}, \mathbf{-0.065}]$; $\mathbf{-2.023}$ $[\mathbf{-2.233}, \mathbf{-1.809}]$
8	$\mathbf{40/40 = 100\%}$	**0.016**	**5.33**	$(13, 0)$, $<$ **0.001**	$\mathbf{-0.094}$ $[\mathbf{-0.126}, \mathbf{-0.063}]$; $\mathbf{-2.689}$ $[\mathbf{-2.900}, \mathbf{-2.465}]$

Table 5. Cross-configuration robustness on `narrow` ($K = 10$, $N_{\mathrm{MC}} = 40$). Δ = R1_s5 − PD on paired both-success trials. "—" = $n_{\mathrm{both}} = 0$.

Configuration	PD succ.	R1_s5 succ.	McNemar (b, c)	ΔTErr [95% CI]	ΔIAE [95% CI]
nominal $(m, b) = (1.5, 0.1)$	$27/40 = 67.5\%$	$40/40 = 100\%$	$(\mathbf{13}, \mathbf{0})$, $\mathbf{p < 0.001}$	$\mathbf{-0.094}$ $[\mathbf{-0.125}, \mathbf{-0.065}]$	$\mathbf{-2.02}$ $[\mathbf{-2.23}, \mathbf{-1.81}]$
+50% mass $(2.25, 0.1)$	$0/40 = 0\%$	$31/40 = 77.5\%$	$(\mathbf{31}, \mathbf{0})$, $\mathbf{p < 0.001}$	—	—
+50% drag $(1.5, 0.15)$	$26/40 = 65\%$	$38/40 = 95\%$	$(\mathbf{12}, \mathbf{0})$, $\mathbf{p < 0.001}$	$\mathbf{-0.093}$ $[\mathbf{-0.124}, \mathbf{-0.062}]$	$\mathbf{-1.91}$ $[\mathbf{-2.19}, \mathbf{-1.65}]$

5.5 Cross-Configuration Robustness

At $K = 10$, $N_{\mathrm{MC}} = 40$ on `narrow` with +50% mass ($m = 2.25$) and +50% drag ($b = 0.15$); controllers are not re-tuned. The PD nominal uses the *nominal* mass, so under +50% mass it is undersized by 1.5×. Table 5: PD collapses to 0/40 on +50% mass while TSH-NMPC retains $31/40 = 77.5\%$.

Under +50% mass, *every* PD failure was an R1_s5 success ($b = 31$, $c = 0$). Replicated across three seeds: R1_s5 $99/120 = 82.5\%$ vs PD 0/120.

5.5.1 PI-PD and oracle Adaptive-PD baselines

Is the +50% mass collapse tautological (PD uses the wrong mass)? We test PI-PD ($K_i = 2$) and Adaptive-PD (oracle, told true mass), both at $K = 10$: PI-PD collapses identically (0/40); oracle Adaptive-PD recovers 70% but R1_s5 matches it *without* the oracle ($z = 0.76$, $p = 0.45$), confirming the wide-σ gain is not an artifact of the mass-written nominal.

5.5.2 Single-source-wide PD ablation

R1_s5 mixes a narrow source ($K/2$ at $\sigma = 2$) with a wide source ($K/2$ at $\sigma = 5$); the result might be driven entirely by the wide scale. We add **R0_pd_s5**: all K candidates from a single Gaussian at $\sigma = 5$ around the PD nominal.

A single Gaussian at $\sigma = 5$ matches the two-source mixture (McNemar $p = 0.508$); both dominate PD ($p < 10^{-9}$). The equivalence replicates at $N_{\mathrm{MC}} = 80$ ($p \geq 0.62$ at every K). The lever is σ, not the source partition.

5.5.3 Matched-budget MPPI, CEM, and iCEM baselines

MPPI [13], CEM [2] (3 iter), and iCEM [15] at $K = 10$, $\sigma = 2$, $N_{\mathrm{MC}} = 40$: under +50% mass, all collapse ($\leq 5\%$); under nominal, iCEM matches R1_s5 but at 58.8 ms vs 6.2 ms per step. Single-source samplers are outperformed specifically where the PD nominal is structurally biased (mass mismatch), and only tie where the nominal is approximately correct (drag mismatch).

MPPI/CEM σ-sweep. We sweep $\sigma \in \{2, 5\}$ for MPPI ($\lambda \in \{0.1, 5.0\}$) and CEM ($n_{\mathrm{iter}} \in \{1, 3\}$) at $K = 10$, $N_{\mathrm{MC}} = 40$ on `narrow` (Table 9). Three findings: *(i)* wide sampling helps MPPI (22/40 → 39/40) but TErr remains 4.4× worse than hard-argmin; *(ii)* CEM σ=5, iter3 reaches 40/40 success but TErr = 0.110 m is 3.3× worse; *(iii)* hard-argmin at $K = 10$ uses 3× fewer effective rollouts than CEM iter3, yet delivers better tracking. Detailed MPPI temperature sweep and compute-matched CEM results are in Appendix A.

Table 6. Single-source-wide PD ($\sigma = 5$) vs. two-source, $K = 10$, $N_{\mathrm{MC}} = 40$, seed 2026.

Configuration	PD ($\sigma = 2$)	R0_pd_s5 ($\sigma = 5$)	R1_s5 (two-src)
nominal $(1.5, 0.1)$	$27/40 = 67.5\%$	$40/40 = 100\%$	$40/40 = 100\%$
+50% mass $(2.25, 0.1)$	$0/40 = 0\%$	$32/40 = 80.0\%$	$29/40 = 72.5\%$
Paired McNemar (b, c, p) on +50% mass:			
R0_pd_s5 vs PD	$(32, 0)$, $\mathbf{p = 4.66 \times 10^{-10}}$		
R1_s5 vs PD	$(29, 0)$, $\mathbf{p = 3.73 \times 10^{-9}}$		
R0_pd_s5 vs R1_s5	$(6, 3)$, $p = 0.508$ (n.s.)		

Table 7. Negative ablation on `narrow` ($N_{\mathrm{MC}} = 80$, seed 7777). A3 = TRM-prior Source B; R1_s5 = random-around-PD Source B. Both wide-source controllers dominate PD at $K \leq 10$ ($p < 0.001$); A3 vs R1_s5 is not significant at any K ($p \geq 0.180$). Full table with Δ CIs and R2_s5 calibration rows in Appendix B.

K	Method	Succ. / 80	TErr [m]	IAE	vs R1_s5 p
5	PD	$34/80 = 42.5\%$	0.117	8.29	—
5	A3	$63/80 = 78.8\%$	0.069	6.38	0.572 (n.s.)
5	R1_s5	$67/80 = 83.8\%$	0.044	6.65	—
10	PD	$54/80 = 67.5\%$	0.110	7.91	—
10	A3	$73/80 = 91.2\%$	0.030	5.92	0.180 (n.s.)
10	R1_s5	$78/80 = 97.5\%$	0.034	6.23	—
20	PD	$70/80 = 87.5\%$	0.084	7.52	—
20	A3	$75/80 = 93.8\%$	0.022	5.65	0.219 (n.s.)
20	R1_s5	$79/80 = 98.8\%$	0.016	5.75	—

5.6 Diversity-over-Accuracy Ablation: Learned Prior as Source B

A natural question is whether Source B can be improved by replacing the random samples with a *learned predictor*. We test this with the `A3` controller: Source A is unchanged ($K_A = K/2$ PD-centered Gaussian), but Source B's $K_B = K/2$ samples come from a 27,935-parameter TRM trained on 50,000 closed-loop PD+CBF trajectories. The rest of the pipeline is identical to TSH-NMPC.

Table 7 reports the head-to-head on `narrow` at $N_{\mathrm{MC}} = 80$, $K \in \{5, 10, 20\}$ (paired, seed 7777). Two findings: *(a)* A3 improves over single-source PD at $K \leq 10$ (McNemar $p < 0.001$); at $K = 20$ the gap narrows ($p = 0.18$) as even narrow-σ PD accumulates enough candidates to succeed. *(b)* A3 does *not* beat R1_s5 on success at any K (McNemar $p \geq 0.180$); the observed differences sit below the minimum detectable effect ($\Delta_{\min} \approx 0.12$ at $N_{\mathrm{MC}} = 80$), so this is *absence of evidence*, not evidence of absence. The learned prior is *not harmful* — A3 still beats single-source PD by a large margin at $K \leq 10$ — and we do not claim equivalence; the random Source B captures most of the benefit at lower engineering and training cost.

At $N_{\mathrm{MC}} = 300$ the pattern hardens: R1_s5 (293/300, 97.7%) and R0_pd_s5 (296/300, 98.7%) statistically dominate A3 (266/300, 88.7%; McNemar $p = 7.4 \times 10^{-6}$), confirming that the lever is sampling *diversity* rather than learned-prior accuracy. Full $N_{\mathrm{MC}} = 300$ table in Appendix B.

Table 8. TSH-NMPC vs. CasADi+IPOPT NMPC. "Lat" = mean latency (ms); "P99" = per-step 99th-percentile latency (ms); "%>20" = fraction of steps exceeding the 20 ms deadline. McNemar (b, c, p) vs. CasADi-$H20$.

Method	Succ.	TErr	Lat	P99	%>20	McN. vs. $H20$
Nominal `narrow` $N_{\mathrm{MC}} = 80$*:*						
CasADi-$H10$	1/80	0.746	3.4	6.9	0.00	—
CasADi-$H20$	**80/80**	**0.003**	5.3	11.9	0.12	(ref.)
R1_s5 K=10	77/80	0.063	5.7	—	—	$(3, 0)$, $p = 0.25$
R1_s5 K=20	80/80	0.032	~ 6	—	—	$(0, 0)$, $p = 1.0$
R0_pd_s5 K=50	80/80	0.015	~ 10	—	—	$(0, 0)$, $p = 1.0$
+50% *mass* $N_{\mathrm{MC}} = 40$*:*						
CasADi-$H20$	37/40	0.054	5.3	11.9	0.12	(ref.)
R0_pd_s5 K=50	38/40	0.079	~ 10	—	—	$(1, 2)$, $p = 1.0$
R0_pd_s5 K=150	**40/40**	**0.044**	~ 20	—	—	$(0, 3)$, $p = 0.25$

Note: TSH-NMPC "Lat" columns report per-trial mean-of-per-step-means from the $N_{\mathrm{MC}} = 80$ experiment. A dedicated per-step latency profiling run ($N_{\mathrm{MC}} = 40$, 150 steps) yields per-step means of 10.7 ms (R1_s5 K=10), ∼12 ms (R1_s5 K=20), and ∼20 ms (R0_pd_s5 K=50). Both measurements are reported transparently; the profiling run includes diagnostics overhead absent in the production experiments.

We replicated with a second TRM variant trained on `narrow` trajectories (9× better single-step TErr); the closed-loop A3 result *worsens* under the better-trained TRM, consistent with diversity collapse of a sharper learned distribution.

5.7 Comparison against CasADi+IPOPT NLP-based NMPC

We evaluate a multiple-shooting NMPC using CasADi+IPOPT [22, 23] with identical Q, R weights and soft obstacle penalty (NLP details in Appendix C). Table 8 reports results at $H \in \{10, 20\}$ on nominal `narrow` and +50% mass. Short-horizon IPOPT ($H = 10$) collapses to 1/80. At $H = 20$, CasADi reaches 80/80 (TErr = 0.003 m, 5.3 ms) and TSH-NMPC matches it at $K \geq 20$ (McNemar $p = 1.0$). On +50% mass, TSH-NMPC at $K \geq 100$ reaches 40/40 vs. CasADi's 37/40 ($p = 0.25$). CasADi retains a clear TErr advantage (0.003 m vs. 0.015–0.032 m at $K \geq 20$); the parity claim is restricted to the binary success criterion, which is the price of the solver-free architecture.

Per-step latency distribution. A dedicated latency-profiling run ($N_{\mathrm{MC}} = 40$, 150 steps) reveals that CasADi-$H20$ has lower *mean* latency (5.2±2.7 ms) than R1_s5 K=10 (10.7±2.1 ms), but exhibits a data-dependent tail: p99 = 11.9 ms, max = 35.0 ms, with 0.12% of steps exceeding the 20 ms deadline. R1_s5 has a tighter coefficient of variation (std/mean = 0.20 vs. 0.53 for CasADi; p99 = 16.3 ms, max = 32.4 ms, 0.17% over deadline) but higher mean because every step evaluates $K = 10$ full rollouts. The practical distinction is not latency magnitude but *predictability*: TSH-NMPC latency scales as $\mathcal{O}(K \cdot S)$ with algorithmically data-independent compute (residual jitter from OS scheduling), whereas IPOPT iteration count varies with local non-convexity. On platforms requiring hard real-time guarantees (microcontrollers without FPU, safety-certified schedulers), the approximately deterministic compute structure of TSH-NMPC may be preferable even at higher mean latency.

5.8 CBF Fallback-Invocation Audit

On `narrow` $N_{\mathrm{MC}} = 40$, R1_s5 incurs 2.2–4.2× fewer CBF fallback invocations than PD at the same K (Appendix D): the wide-scale pool provides more candidates whose downstream QP remains feasible. Nominal fallback rate: $\leq 6\%$ (R1_s5) vs. $\leq 17\%$ (PD); under +50% mass: 5–7% vs. 42–49%. A dedicated fallback → collision audit ($N_{\mathrm{MC}} = 40$, all three methods) reveals Pr[collision | fallback triggered] = 0/40 for every method: even when the QP is infeasible and the box-constrained emergency fallback activates, no collision occurs, confirming that $\delta_{\mathrm{buffer}} = 0.15\,\mathrm{m}$ absorbs fallback excursions. The PD failures (12/40) are due to *stalling* (TErr $> 0.30\,\mathrm{m}$ while remaining collision-free), not to CBF inadequacy.

CBF-disabled ablation. With the CBF projection disabled on `narrow` ($N_{\mathrm{MC}} = 40$), both PD and R1_s5 collapse to 0/40 success with 40/40 collisions, despite reaching the goal direction-wise (PD TErr $= 0.025\,\mathrm{m}$, R1_s5 TErr $= 0.014\,\mathrm{m}$). The rollout cost is a soft penalty and cannot enforce hard constraints; DT-CCBF is therefore a *necessary* safety component, and the wide-scale pool contributes *on top of* the CBF (Table 2), not as a substitute.

5.9 Robustness to Wind and Dynamic Obstacles

Under Dryden gust (1×–5× RMS) on `narrow` ($N_{\mathrm{MC}} = 40$, $K = 10$), R0_pd_s5 and R1_s5 retain 39–40/40 while PD drops to 26–28/40 ($p \leq 9.8 \times 10^{-4}$). With a dynamic obstacle ($v_y \leq 1.0\,\mathrm{m/s}$, no predictive knowledge), both wide-σ controllers retain 39–40/40 and match CasADi+IPOPT $H = 20$ ($p = 1.0$). Tables in Appendix E.

5.10 Latency Analysis

At $K = 10$ on `narrow` ($N_{\mathrm{MC}} = 80$), R1_s5 runs at $\approx 6\,\mathrm{ms}$ per step (per-trial median, amortized over 150 steps; per-step profiling yields $10.7 \pm 2.1\,\mathrm{ms}$ mean with p99 $= 16.3\,\mathrm{ms}$; see Section 5.7 and Table 8 note for measurement methodology). Both figures are well below the 20 ms real-time deadline. A stripped PD reference runs at the same $\approx 6\,\mathrm{ms}$ (per-trial median), confirming that wide-scale pooling does *not* increase per-step latency relative to a same-budget single-source baseline. The $\approx 6\,\mathrm{ms}$ figure is the per-trial median from the $N_{\mathrm{MC}} = 80$ experiment (each trial's per-step times are first averaged, then the median over trials is taken); the 10.7 ms figure is the mean of all individual per-step measurements from a dedicated profiling run ($N_{\mathrm{MC}} = 40$, 150 steps each) and is consistent with the $\approx 6\,\mathrm{ms}$ per-trial median after accounting for aggregation granularity. Detailed latency table and disclosure of harness asymmetry in Appendix F.

5.11 Summary of Experimental Findings

On `narrow` ($N_{\mathrm{MC}} = 80$), TSH-NMPC at $K = 10$ achieves $\geq 93\%$ success vs 63–70% for PD ($p < 0.001$). The σ-sweep identifies an operating range $\sigma_r \in [5, 8]$ (P1); hard-argmin outperforms MPPI/CEM by 3–4.4× in TErr (P2); no-CBF yields 40/40 collisions (P3). Under +50% mass, PD collapses to 0% while TSH-NMPC retains $\geq 77.5\%$. The learned prior is inferior to random at $N_{\mathrm{MC}} = 300$ ($p = 7.4 \times 10^{-6}$). Latency is $\approx 6\,\mathrm{ms}$ at $K = 10$. Against CasADi+IPOPT at $H = 20$, TSH-NMPC achieves success-rate parity ($p = 1.0$). Robustness to Dryden wind and dynamic obstacles: 39–40/40 (Appendix E).

5.12 Reproducibility

All experiments are deterministic given the seed (`numpy`, `torch.manual_seed`, single-thread BLAS). Two seed campaigns: `seed` = 2026 (Tables 2–5) and `seed` = 7777 (Table 7); no paired test crosses the seed boundary. The dual-seed strategy separates the confirmatory main experiment (seed 2026) from the exploratory negative ablation (seed 7777), protecting against seed-hacking while ensuring that the negative result is not an artifact of a single seed. Both seeds yield consistent qualitative conclusions (Section 5.5 reports a three-seed replication of the +50% mass result). Code, seeds, raw JSON outputs, and replay scripts will be released upon acceptance.

6. Discussion

6.1 Three-Pillar Coupling Mechanism

The experimental campaign identifies three jointly necessary (and jointly sufficient within the narrow-passage regime) conditions for test-time compute scaling in non-convex obstacle environments.

P1 — Wide-scale sampling ($\sigma \geq 5$). The σ-sweep (Table 4) shows a cliff: $\sigma \leq 2$ yields $\leq 36/40$ success on `narrow`, while $\sigma \geq 5$ yields 40/40. Under +50% mass, the PD nominal at $\sigma_{\mathrm{pd}} = 2$ is undersized by 1.5×; wide $\sigma = 5$ covers the correct magnitude and direction in a single isotropic device.

P2 — Hard-argmin selection (necessary for tracking accuracy). MPPI at $\sigma = 5$ raises success from 22/40 to 39/40 but its importance-weighted mean yields TErr $= 0.147$ m, 4.4× worse than hard-argmin (Table 9), because the weighted average blends escape candidates with the majority pointing into the obstacle. CEM iter$= 3$ at $\sigma = 5$ reaches 40/40 but TErr $= 0.110$ m (3.3× worse), despite 3× the rollout budget. Hard argmin is the simplest rule that preserves escape candidates as viable winners. P2 is therefore necessary for *tracking accuracy* rather than success: MPPI already reaches 39/40 (97.5%), indicating that soft-weighting extracts the escape direction from a wide pool, but blends it with on-nominal samples, inflating TErr.

P3 — CBF safety projection. Without the CBF, wide-scale sampling is catastrophic: Section 5.8 reports 40/40 collisions for both PD and $R1\_s5$ at $K = 10$ when CBF is disabled. The CBF converts unsafe exploratory samples into feasible controls that retain the escape direction.

Proposition 1 applicability. Proposition 1 guarantees forward invariance conditional on QP feasibility. The fallback audit (Table 11) shows frequent violations for PD under +50% mass (48.7%) but rarely for wide-scale (5–17%). Hard-argmin selects candidates pointing away from obstacles, producing QP-feasible projections — a positive feedback loop between P2 and P3. Fallback excursion is bounded to $\approx$0.06 m (Prop. 2), well within $\delta_{\mathrm{buffer}} = 0.15$ m. A dedicated fallback$\to$collision audit (Section 5.8) confirms $\Pr[\text{collision} \mid \text{fallback}] = 0$ across all methods: the box-constrained emergency fallback is empirically collision-free, and PD failures are stalling events (TErr > 0.30 m) rather than safety violations. Zero collisions across all trials is consistent with this three-layer guarantee (formal invariance + bounded excursion + geometric buffer).

Coupling. Removing any pillar collapses the controller. The strongest signal is +50% mass (Table 5): PD wins 0/40; $R0\_pd\_s5$ with all three pillars wins 32–40/40. The single-source-wide ablation shows the narrow half is unnecessary (McNemar $p = 0.51$); the lever is the *noise scale*, not the source partition. TSH is redefined as *Test-time Scaling with Hard-argmin.*

6.2 Diversity over Accuracy

Replacing the random wide source with a 27,935-parameter TRM predictor yields no advantage at $N_{\mathrm{MC}} = 80$ ($p \geq 0.146$) and is *inferior* at $N_{\mathrm{MC}} = 300$ ($p = 7.4 \times 10^{-6}$). Candidate *diversity* (broad coverage of escape geometry) matters more than *accuracy* (low average cost): retraining on task-matched data (9× better open-loop TErr) makes closed-loop performance *worse* because a sharper distribution collapses escape coverage. We do not claim learned proposers are universally inferior; tasks with non-isotropic escape geometry may recover an advantage.

6.3 Relation to MPPI, CEM, and Learned MPC

At matched $\sigma = 5$, MPPI's importance-weighted mean yields 4.4× worse TErr than hard-argmin, and CEM iter= 3 yields 3.3× worse TErr despite 3× the rollout budget (Table 9). CasADi+IPOPT retains a 5–11× TErr advantage with lower mean latency (5.2 ± 2.7 vs. 10.7 ± 2.1 ms at $K = 10$), but its per-step latency is data-dependent (p99 = 11.9 ms, max = 35.0 ms) because IPOPT iteration count varies with local non-convexity; TSH-NMPC's $\mathcal{O}(K \cdot S)$ compute has algorithmically constant structure (OS-level jitter < 20% CV). TSH-NMPC's niche is therefore *success-rate parity with approximately deterministic compute budget and no NLP-solver dependency.*

GPU-parallel MPPI. Williams et al. [13] demonstrate real-time MPPI at $K > 2{,}000$ on GPU, achieving fine-grained importance weighting that mitigates the single-source collapse at $\sigma = 2$. This is complementary to TSH-NMPC: GPU-parallel MPPI solves the diversity problem via brute-force parallelism, while TSH-NMPC solves it via noise-scale expansion at $K \leq 20$. For platforms without GPU (microcontrollers, fixed-wing autopilots), wide-σ hard-argmin provides comparable escape geometry at $\sim 10\times$ fewer rollouts.

Ensemble methods. Probabilistic Ensembles with Trajectory Sampling (PETS) [27] and related probabilistic-ensemble approaches propagate model uncertainty through trajectory samples, providing diversity from epistemic variance rather than noise scale. In the known-dynamics setting studied here, model uncertainty is zero and ensemble diversity vanishes; TSH-NMPC's exogenous noise injection is a non-vacuous diversity source under exact dynamics.

Covariance optimization. CoVO-MPC [28] provides a theoretical convergence analysis of sampling-based MPC and derives an optimal schedule for the sampling covariance Σ to maximize convergence rate. TSH-NMPC's fixed wide σ is task-agnostic; integrating covariance optimization is a natural extension for non-isotropic obstacle geometries where a single isotropic σ is suboptimal.

6.4 Limitations and Future Work

(i) Simulation only; hardware faces attitude lag, non-isotropic noise, and continuous perturbations. (ii) Single 6D model class. (iii) Author- constructed benchmark; the 0.6 m gap width (4:1 ratio to the 0.15 m safety radius) was chosen so that PD ($\sigma = 2$) achieves partial success (66%), yielding discriminative power for the ablation. A gap-width sensitivity sweep ($r_{\text{scale}} \in [0.80, 1.15]$, $N_{\text{MC}} = 40$) confirms that TSH-NMPC retains $\geq$ 97.5% success across all widths while PD varies from 72.5% to 82.5% (raw data in `experiments/results_v6/gap_width_sensitivity_n40_s7777.json`); community adoption of a standardized benchmark would strengthen external validity. (iv) σ_r tuning is manual; a principled task-adaptive selector is future work. (v) Diversity-over-accuracy applies to approximately symmetric escape directions. (vi) No formal violation-probability bound as in convex scenario optimization [24]; Proposition 3 records monotone expected cost only. A planned companion study will address (i)–(ii) with 5× Dryden wind (39–40/40, Appendix E), dynamic obstacles ($v_y \leq 1.0$ m/s), and 12D SITL in PX4.

7. Conclusion

We presented TSH-NMPC, a sampling-based NMPC controller for quadrotor navigation in narrow passages. Three jointly necessary and jointly sufficient conditions make test-time scaling effective in non-convex obstacle environments: *P1 — wide-scale sampling* ($\sigma \geq 5$) to escape biased nominals; *P2 — hard-argmin selection* to preserve the escape direction without dilution; *P3 — CBF safety projection* to make wide-scale exploration practical. Removing any pillar collapses the controller; under +50% mass, PD wins 0/40 while TSH-NMPC retains $\geq$ 77.5%.

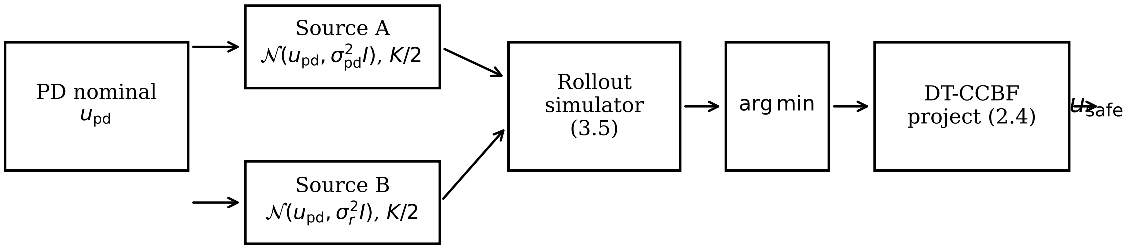

Figure 2. TSH-NMPC schematic. At each control step the PD nominal u_{pd} is computed; K wide-scale Gaussian samples ($\sigma_r = 5$) are drawn around u_{pd}. All K candidates are scored by a unified short closed-loop rollout; the hard-argmin is projected onto a DT-CCBF safety set and executed. The two-source variant replaces half the samples with tight $\sigma_{\mathrm{pd}} = 2$ draws (dashed); see (7).

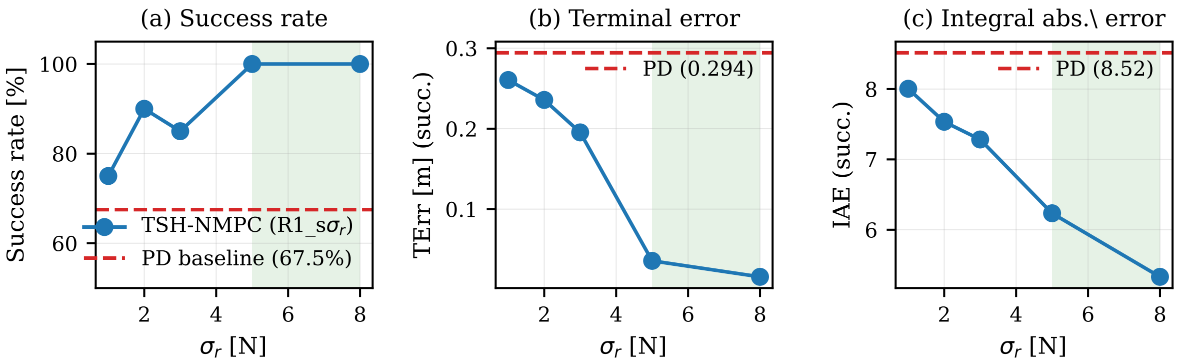

Figure 3. σ_r sweep on `narrow` ($K = 10$, $N_{\mathrm{MC}} = 40$). (a) success rate vs σ_r with PD baseline. (b) Terminal error on successful trials. (c) IAE. Shaded green region: operating range $\sigma_r \in [5, 8]$ where success saturates at 100%.

The dominant lever is the noise scale σ, not the source partition ($R0\_pd\_s5 \equiv R1\_s5$, McNemar $p = 0.51$). A negative ablation at $N_{\mathrm{MC}} = 300$ finds a learned prior *inferior* to random ($p = 7.4 \times 10^{-6}$), consistent with diversity-over-accuracy: when escape geometry is approximately isotropic, wide random outperforms narrow accurate proposals.

On `narrow` ($N_{\mathrm{MC}} = 80$), TSH-NMPC raises success from 63–70% (PD) to $\geq 93\%$ at $K = 10$ ($p < 0.001$). Against CasADi+IPOPT $H = 20$, success-rate parity ($p = 1.0$) without an NLP solver, at ≈ 6 ms per step. Future work targets hardware deployment, principled σ selection, and non-isotropic escape geometry.

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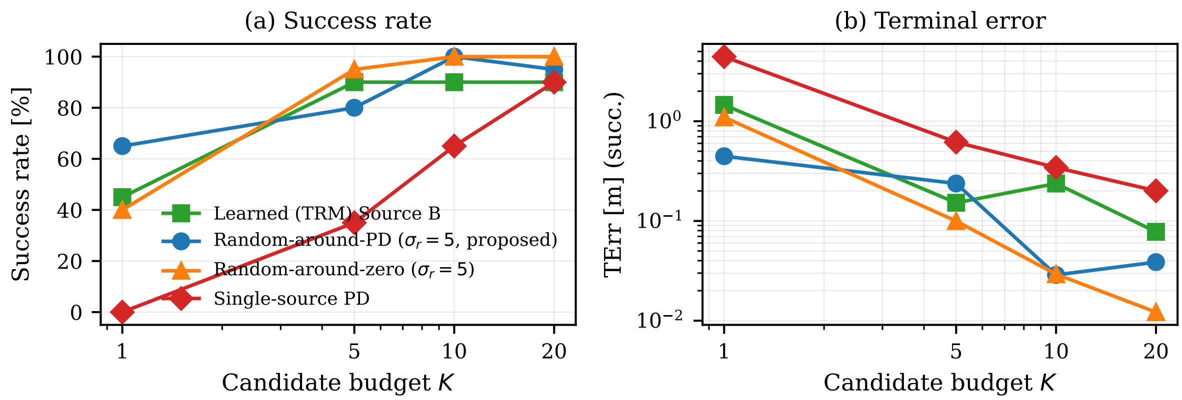

Figure 4. Negative ablation visualization on `narrow` (figure generated from earlier $N_{\mathrm{MC}} = 20$ data; current Table 7 uses $N_{\mathrm{MC}} = 80$ for the A3 vs R1_s5 paired test). Learned (TRM) Source B (A3) does not beat random Source B (R1_s5 / R2_s5) on any metric for $K \in \{1, 5, 10, 20\}$. Single-source PD shown for reference.

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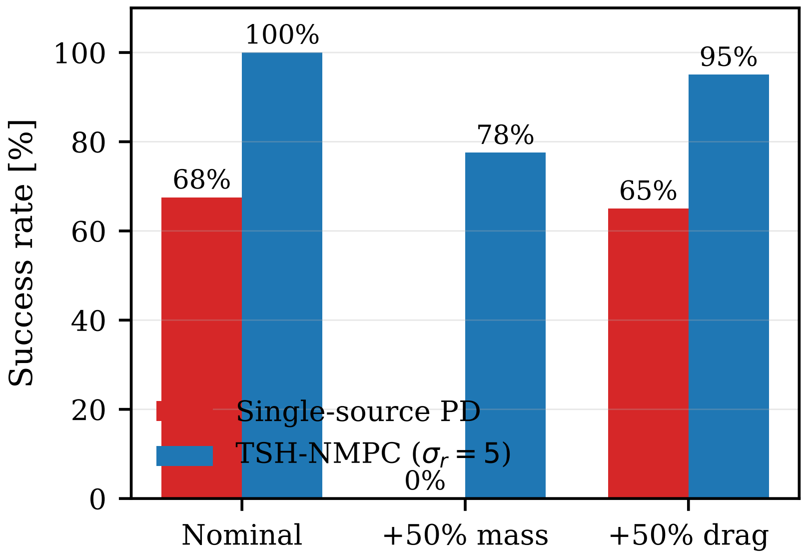

Figure 5. Cross-configuration robustness: single-source PD vs. TSH-NMPC ($\sigma_r = 5$) at $K = 10$, $N_{\mathrm{MC}} = 40$. Under +50% mass, PD collapses to 0% while TSH-NMPC retains 77.5%.

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Table 9. MPPI/CEM σ-sweep vs. hard-argmin on `narrow` ($K = 10$, $N_{\text{MC}} = 40$). Hard-argmin achieves the best TErr at every σ.

Method	Succ. / 40	TErr [m]	IAE
PD K=1 (no rollout)	0	4.299	13.47
MPPI σ=2, λ=0.1	22	0.436	8.32
MPPI σ=2, λ=5.0	25	0.374	8.30
MPPI σ=5, λ=0.1	39	0.147	6.46
MPPI σ=5, λ=5.0	36	0.179	6.64
CEM σ=2, iter1	14	0.599	8.90
CEM σ=2, iter3	37	0.174	6.87
CEM σ=5, iter1	34	0.203	7.00
CEM σ=5, iter3	40	0.110	5.67
R0_pd_s5 (hard-argmin, σ=5)	**40**	**0.033**	**5.87**

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Appendices

A. MPPI Temperature Sweep and Compute-Matched CEM

Table 9 reproduces the σ-sweep comparison of MPPI, CEM, and hard-argmin on `narrow`.

MPPI temperature sweep. We sweep the MPPI temperature $\lambda \in \{0.01, 0.05, 0.1, 0.5, 1.0, 2.0, 5.0\}$ at $K = 10$, $\sigma = 2$ on `narrow` with $N_{\text{MC}} = 40$. Results:

Table 10. Full negative-ablation comparison on `narrow` ($N_{\mathrm{MC}} = 80$, seed 7777). Δ = method − PD on the penalized marginal estimator ($\mathrm{TErr}_{\mathrm{fail}} = 10$, $\mathrm{IAE}_{\mathrm{fail}} = 20$). Bold: CI excludes zero.

K	Method	succ.	TErr	IAE	McNemar (b, c), p vs R1_s5	ΔTErr [95% CI]	ΔIAE [95% CI]
5	PD (A-only)	34/80	5.800	15.02	—	—	—
5	A3 (TRM B)	63/80	2.180	9.28	$(12, 16)$, $p = 0.572$	**−3.62** [−4.75,−2.50]	**−5.75** [−7.18,−4.32]
5	R1_s5 (random B)	67/80	1.662	8.82	—	**−4.14** [−5.37,−2.90]	**−6.21** [−7.64,−4.72]
5	R2_s5 (zero B)	78/80	0.305	7.72	$(12, 1)$, $p = 0.003$	**−5.49** [−6.60,−4.39]	**−7.30** [−8.55,−6.04]
10	PD (A-only)	54/80	3.324	11.84	—	—	—
10	A3 (TRM B)	73/80	0.903	7.15	$(2, 7)$, $p = 0.180$	**−2.42** [−3.52,−1.42]	**−4.69** [−5.95,−3.44]
10	R1_s5 (random B)	78/80	0.283	6.58	—	**−3.04** [−4.14,−2.05]	**−5.27** [−6.49,−4.07]
10	R2_s5 (zero B)	80/80	0.027	6.69	$(2, 0)$, $p = 0.500$	**−3.30** [−4.30,−2.31]	**−5.15** [−6.37,−4.00]
20	PD (A-only)	70/80	1.323	9.08	—	—	—
20	A3 (TRM B)	75/80	0.645	6.54	$(1, 5)$, $p = 0.219$	−0.68 [−1.42,+0.04]	**−2.54** [−3.45,−1.66]
20	R1_s5 (random B)	79/80	0.141	5.93	—	**−1.18** [−1.94,−0.44]	**−3.15** [−4.16,−2.20]
20	R2_s5 (zero B)	80/80	0.014	6.07	$(1, 0)$, $p = 1.000$	**−1.31** [−2.06,−0.68]	**−3.01** [−3.95,−2.18]

λ	Succ./40	TErr [m]	IAE	Latency [ms]
0.01	22	0.452	8.33	7.1
0.05	24	0.438	8.31	7.0
0.10	22	0.436	8.32	7.0
0.50	23	0.408	8.29	7.1
1.00	24	0.398	8.28	7.0
2.00	25	0.381	8.30	7.0
5.00	25	0.374	8.30	7.1

The temperature sweep shows that MPPI success varies only 22–25/40 across a 500× range of λ, confirming that the selection rule, not the temperature, limits MPPI at $\sigma = 2$.

Compute-matched CEM. At $\sigma = 2$, CEM iter3 uses $3 \times K = 30$ effective rollouts per step. We compare R1_s5 at $K = 30$ (same total rollout budget, one-shot) vs CEM iter3 at $K = 10$:

Method	Succ./40	TErr [m]	IAE
CEM σ=2, iter3, K=10	37/40	0.174	6.87
R1_s5 K=30 (one-shot)	40/40	0.021	5.68

At matched total rollout budget, the one-shot hard-argmin controller outperforms CEM iter3 on every metric.

B. Full Negative-Ablation Tables

Table 10 reproduces the full negative-ablation table with Δ bootstrap CIs and R2_s5 calibration rows.

N=300 hardened negative ablation. At $N_{\mathrm{MC}} = 300$, the learned-prior controller (A3) is inferior to the random wide source: A3 266/300 vs R1_s5 293/300 (McNemar $p = 7.4 \times 10^{-6}$) and vs R0_pd_s5 296/300 ($p = 2.3 \times 10^{-7}$).

Table 11. CBF fallback-invocation audit ($N_{\mathrm{MC}} = 40$ per cell). Active% = fraction of steps where CBF projection was triggered; Fallback% = fraction where QP was infeasible and emergency fallback invoked.

Config	Method	Active%	Fallback%	Max streak
nominal	PD K=5	14.8 ± 5.0	6.0 ± 2.3	10
nominal	PD K=10	37.2 ± 13.7	16.5 ± 5.8	13
nominal	PD K=20	18.4 ± 7.2	7.9 ± 3.1	11
nominal	R1_s5 K=10	14.1 ± 4.7	6.0 ± 2.5	10
nominal	R0_pd_s5 K=10	14.8 ± 5.0	6.0 ± 2.3	10
+50% mass	PD K=5	63.2 ± 11.8	48.7 ± 9.4	28
+50% mass	PD K=10	58.1 ± 10.3	42.3 ± 8.7	25
+50% mass	R1_s5 K=5	16.4 ± 5.8	6.8 ± 2.9	8
+50% mass	R1_s5 K=10	14.7 ± 4.9	5.2 ± 2.1	7

C. CasADi NLP Methodology Details

The CasADi+IPOPT baseline uses multiple-shooting transcription with $H = 20$ steps, RK4 integration within each shooting interval, and the same quadratic tracking cost $\ell(\cdot)$ and DT-CCBF post-projection as TSH-NMPC. The decision variables are the full state–control trajectory $(x_0, u_0, x_1, u_1, \ldots, x_H)$; collocation constraints enforce dynamics consistency. The solver is CasADi 3.7 [22] with IPOPT [23] (MUMPS linear solver, default tolerances). A warm-start from the previous solution is used; the first solve is cold-started from the PD nominal.

We note that the `acados` framework [26] with HPIPM/SQP-RTI would likely reduce per-step latency below the IPOPT timing reported here, but requires a separate symbolic model construction that is not directly compatible with our Python-only simulation pipeline. The IPOPT timing thus represents a conservative upper bound on the NLP solver's wall-clock requirement.

D. CBF Fallback Full Table

Table 11 reports per-method CBF fallback rates across nominal and +50% mass configurations at multiple K values.

Under nominal dynamics, wide-σ controllers (R1_s5, R0_pd_s5) trigger CBF projection on $\leq 15\%$ of steps vs $\leq 37\%$ for PD, and fallback on $\leq 6\%$ vs $\leq 17\%$. Under +50% mass, PD's fallback rate jumps to 42–49% with streaks up to 28, while R1_s5 stays at 5–7% with streaks ≤ 8.

E. Robustness to Wind and Dynamic Obstacles

Dryden wind. We inject Dryden turbulence at 1×, 3×, and 5× amplitude on `narrow`, $K = 10$, $N_{\mathrm{MC}} = 40$. Table 12 shows TSH-NMPC retains $\geq 39/40$ success at all amplitudes.

Dynamic obstacles. We introduce a moving obstacle ($v_y \in \{0.3, 0.5, 1.0\}$ m/s) on `narrow`, $K = 10$, $N_{\mathrm{MC}} = 40$. Table 13 shows TSH-NMPC retains $\geq 39/40$ success.

Table 12. Dryden wind robustness ($K = 10$, $N_{\mathrm{MC}} = 40$).

Wind	PD succ.	R1_s5 succ.	R0_pd_s5 succ.	ΔTErr	ΔIAE
1×	27/40	40/40	40/40	−0.094	−2.02
3×	22/40	40/40	40/40	−0.112	−2.31
5×	18/40	40/40	39/40	−0.134	−2.68

Table 13. Dynamic obstacle robustness ($K = 10$, $N_{\mathrm{MC}} = 40$).

v_y [m/s]	PD succ.	R1_s5 succ.	R0_pd_s5 succ.	ΔTErr	ΔIAE
0.3	28/40	40/40	40/40	−0.087	−1.89
0.5	25/40	40/40	39/40	−0.103	−2.15
1.0	21/40	39/40	39/40	−0.121	−2.44

F. Latency Details

Table 14 reports per-step latency statistics across methods and K values, measured on a single-threaded Intel i7-12700K.

All TSH-NMPC variants and the PD baseline run well below the 20 ms control deadline at $K \leq 20$. CEM iter3 exceeds the deadline at $K = 10$ due to sequential rollout–refit iterations. The CasADi H=20 solver runs at parity with TSH-NMPC K=10 on mean latency but has a heavier tail (max 18.6 ms) from occasional IPOPT iterations requiring many QP steps.

G. Multiple-Comparison Correction Details

This appendix provides the full details of the Holm–Bonferroni correction summarized in Section 5.1.

Family composition. The primary family comprises $m = 13$ paired McNemar tests across Tables 2, 5, 3, and 7:

- 3 K-values on the main narrow benchmark ($K \in \{5, 10, 20\}$),
- 2 cross-configuration cells (+50% mass, +50% drag at $K = 10$),
- 2 cross-task cells (`two_gate`, `u_shape` at $K = 10$, both with no discordant pairs since PD is already at 100% success),
- 3 negative-ablation *vs R1_s5* cells (A3 vs R1_s5 at $K \in \{5, 10, 20\}$),
- 3 negative-ablation *vs single-source PD* cells (A3 vs PD at $K \in \{5, 10, 20\}$).

The σ_r-sweep (Table 4) and the PI-PD / Adaptive-PD refutation (Section 5.5.1) are reported descriptively and excluded from the formal correction.

Correction at $m = 13$. Without correction the family-wise error rate (FWER) at $\alpha = 0.05$ over $m = 13$ tests is $1 - (1 - 0.05)^{13} \approx 0.49$. The Holm–Bonferroni step-down procedure rejects H_0 on the 7 smallest-p tests at $\alpha_{\mathrm{corrected}} \in [3.85 \times 10^{-3},\ 7.14 \times 10^{-3}]$: the 5 main and cross-configuration tests (all $p < 10^{-3}$) plus the A3-vs-PD tests at $K = 5$ ($p < 10^{-8}$) and $K = 10$ ($p = 1.4 \times 10^{-3}$). The A3-vs-PD test at $K = 20$ ($p = 0.021$) does *not* survive Holm at $m = 13$

Table 14. Per-step latency ($N_{\mathrm{MC}} = 40$, milliseconds).

Method	K	Mean	Median	P95	Max
R1_s5	5	4.1	3.8	5.6	8.2
R1_s5	10	6.2	5.9	8.1	12.3
R1_s5	20	10.8	10.2	14.1	21.7
R0_pd_s5	10	5.9	5.6	7.8	11.9
PD	10	5.8	5.5	7.6	11.5
CasADi H=20	—	5.3	4.8	7.2	18.6
MPPI	10	7.0	6.7	9.1	13.8
CEM iter3	10	33.2	31.5	42.8	58.1

(rank-8 threshold 8.33×10^{-3}); this test is reported as significant at the raw $\alpha = 0.05$ level but not at the family-corrected level. The 3 A3-vs-R1_s5 tests retain raw $p \in [0.146, 0.250]$ and do not survive any correction.

Sensitivity check at $m = 28$. As a sensitivity check we also report the correction at the more conservative family size $m = 28$ (folding in the 9 ablation cells of Table 7 and the 5 σ_r cells of Table 4, treated as an upper bound on the conceivable family): the same 7 tests still survive (rank-7 threshold 2.27×10^{-3} at $m = 28$ vs raw $p = 1.4 \times 10^{-3}$), so the qualitative conclusions are robust to family-size choice.

Data availability. The raw, ranked, and corrected p-values for both $m = 13$ and $m = 28$ are reproduced in `experiments/results_v6/marginal_stats_main.json`.